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Altman Z-Score & Piotroski F-Score

Financial-strength factors and the distress anomaly

Evidence from the Global Factor Data (Jensen-Kelly-Pedersen, 2023)

Mirko Balli · Lorenzo Attanasio

Group 25 · 5376 Asset / Risk Management I
WU Vienna · 2026



Two Lenses on Financial Strength

From the Global Factor Data we pick two fundamentals-based **quality** factors. The contrarian instinct says: buy distressed firms and profit from their recovery. We test whether the market actually rewards that bet.

Research question

Is financial strength priced in the cross-section, and does betting on distress pay off?

Both sit in the JKP **Quality** theme: Z proxies 'safety', F 'fundamental strength': the pillars of Quality-minus-Junk (Asness et al., 2019).

Altman Z-Score 1968

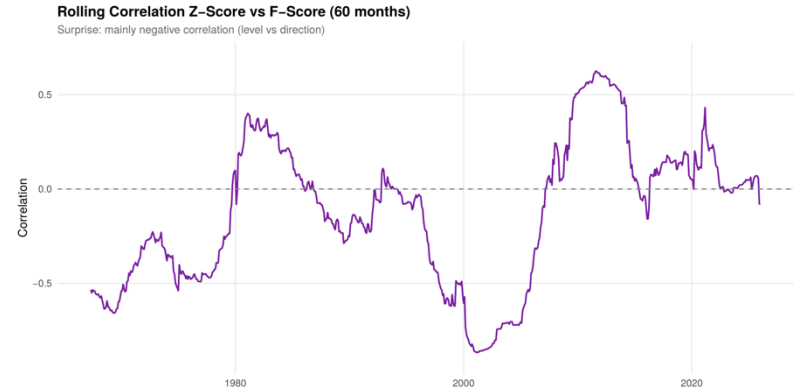
Distance to bankruptcy from five balance-sheet ratios. Captures the LEVEL of health: how safe is the firm today?

Piotroski F-Score 2000

Nine binary accounting signals (score 0-9). Captures the DIRECTION of fundamentals: is the firm improving?

Level vs. Direction, and the Distress Anomaly

- Both factors are long-short: long high-score (safe / improving) firms, short low-score (distressed / weak).
- Contrarian theory: distressed firms are riskier, so they ought to earn a premium.
- The distress anomaly says the opposite: distressed stocks systematically underperform (Campbell-Hilscher-Szilagyi 2008; Dichev 1998).
- So, the real edge is not 'cheap' but 'improving', exactly what Piotroski (2000) sorts on.



Rolling 60-month correlation: mostly negative over time

Key correlations (full sample)

Z-Score vs F-Score: **-0.36**

Z-Score vs Quality: **+0.56**

F-Score vs Market: **-0.37**

F is defensive; Z aligns with the Quality theme

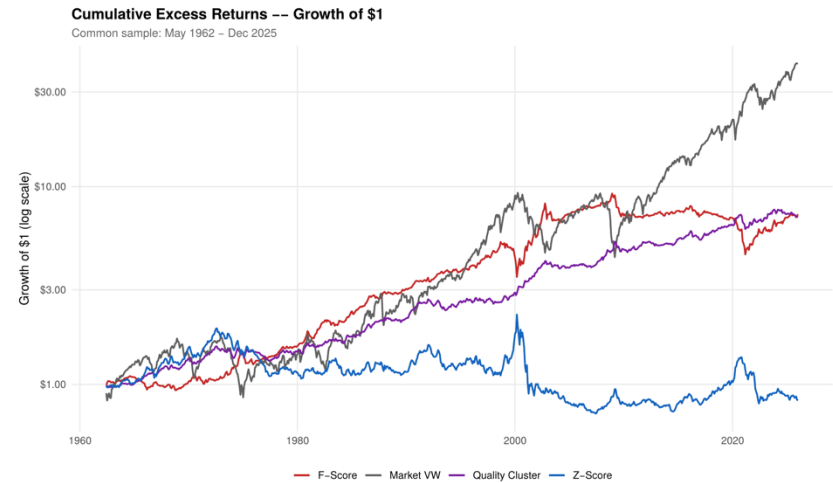
Risk, Return and Significance



Factor	Ret	Vol	SR	t
Z-Score	0.32%	10.99%	0.03	0.21
F-Score	3.23%	7.52%	0.43	3.28
Quality Cluster	2.55%	12.28%	0.21	2.88
Market VW	8.22%	18.38%	0.45	4.32
Market EW	11.26%	25.12%	0.45	4.05

Annualised, full sample. t = Newey-West. Quality skewed by a 1932 outlier; post-1962 Sharpe 0.60 (as charted).

Only F-Score clears the Harvey-Liu-Zhu hurdle of $t > 3$ ($t = 3.28$). Z-Score is indistinguishable from zero.



Growth of \$1, excess returns (log scale), common sample from 1962

Long Leg vs. Short Leg

Each factor is split into its safe/strong leg (top tercile) and distressed/weak leg (bottom tercile) to see where returns originate.

Z-Score: decomposition long Leg vs short Leg

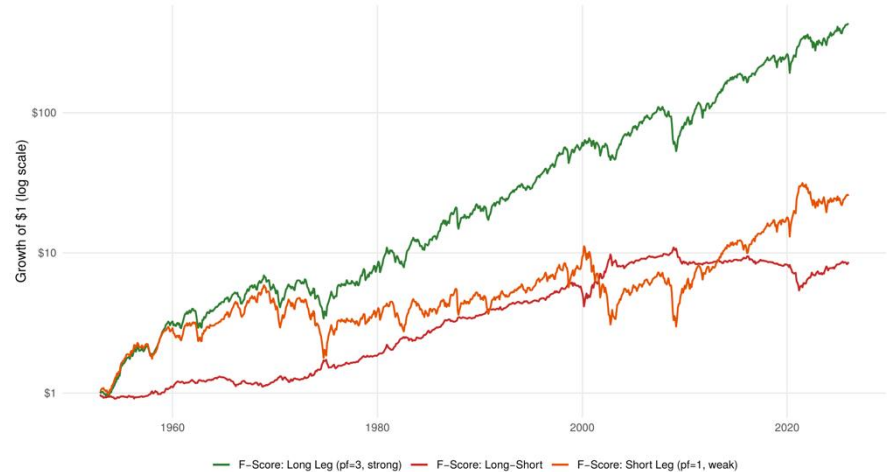
Both legs grow alike: distressed do NOT underperform -> no Z premium



Z-Score: both legs grow alike -> no premium

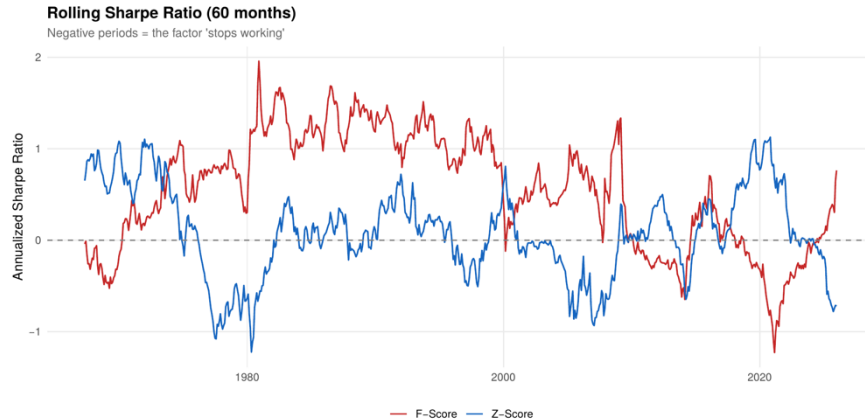
F-Score: decomposition long Leg vs short Leg

Wide gap: strong fundamentals beat weak -> real F premium

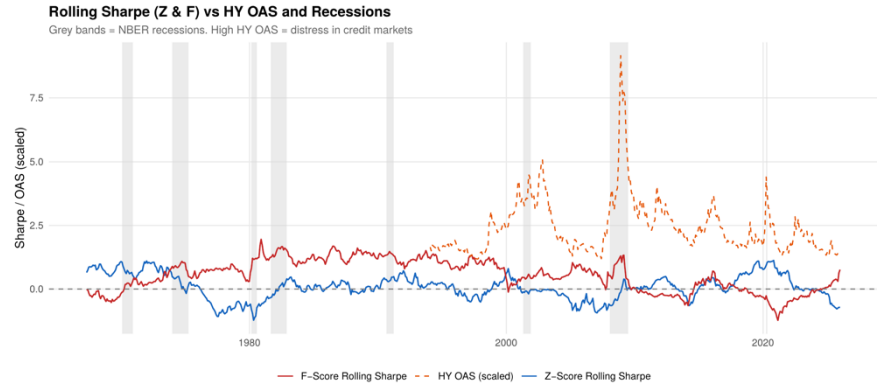


F-Score: strong clearly beats weak -> real premium

Rolling Performance Through Regimes



60-month rolling Sharpe: long flat / negative stretches



Rolling Sharpe vs HY credit spread & NBER recessions

Factor premia are time-varying, not constant: both factors weaken for years at a time, a key risk-management caveat for the strategy.

Performance Before & After Publication

Factor	Period	Ret	SR	t
Z-Score	In-sample ≤ 1965	4.05%	0.80	1.48
Z-Score	Pre-pub 1966-67	7.07%	0.77	1.10
Z-Score	Post-pub ≥ 1968	-0.15%	-0.01	-0.09
F-Score	In-sample ≤ 1996	3.83%	0.74	4.48
F-Score	Pre-pub 1997-99	0.91%	0.10	0.18
F-Score	Post-pub ≥ 2000	2.48%	0.24	1.18

*Annualised. 'Pre-pub' = after the paper's sample, before publication.
Z window is short (data start 1962).*

F-Score

t-stat falls 4.48 $\rightarrow$ 1.18 after publication;
the classic McLean-Pontiff (2016) decay.

Z-Score

Never significant out-of-sample; the in-sample strength did not survive.

Do Standard Models Price Them?

Factor	Model	alpha	t	R ²
Z-Score	CAPM	-0.9%	-0.50	6%
Z-Score	FF3	+2.7%	2.84	64%
Z-Score	q5	+1.8%	0.95	41%
Z-Score	FF6	+3.1%	2.99	68%
F-Score	CAPM	+4.7%	4.87	14%
F-Score	FF3	+4.4%	4.87	29%
F-Score	q5	-0.8%	-1.04	61%
F-Score	FF6	+1.2%	1.68	57%

FF5 omitted: FF6 dominates and tells the same story

Alpha annualised, t = Newey-West. q5 = Hou, Mo, Xue & Zhang (2021).

GRS joint test

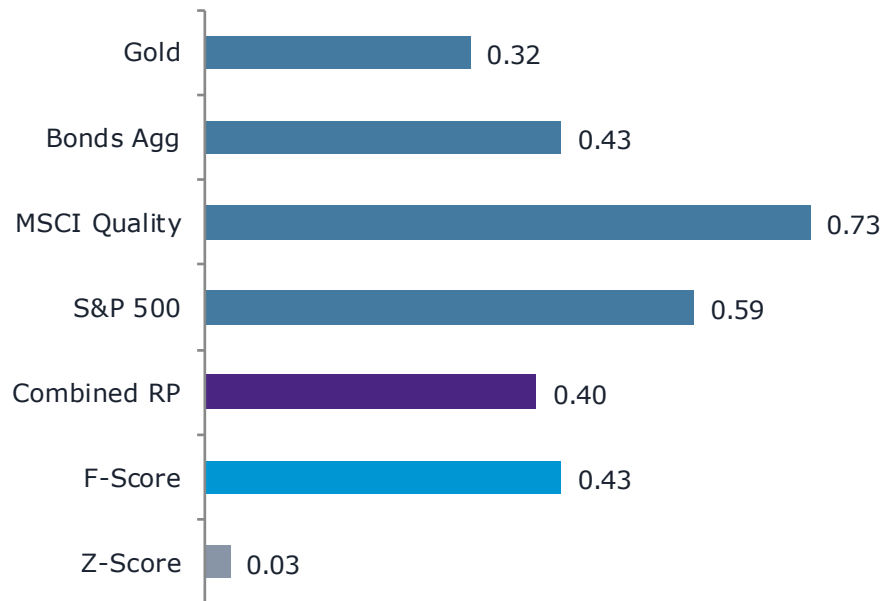
H0: both factors' alphas = 0;
 run per model on { Z-Score , F-Score }:
 CAPM, FF3, FF5, FF6 rejected ($p < 0.01$)
 q5 $p = 0.32$ -> NOT rejected
 Only q5 prices both factors jointly.

F-Score's premium is absorbed by q5 profitability (ROE beta = 0.52, $t = 9.5$).

Z-Score loads strongly negative on value (HML beta = -0.78): a growth / safety bet.

Verdict: Is Financial Strength Priced?

Sharpe ratio: excess returns, annualized, from 1989



1 F-Score is a genuine premium ($t = 3.3$); Z-Score is not ($t = 0.2$); the distress anomaly in action.

2 Level vs direction: the two factors are negatively correlated (-0.36).

3 F-Score's edge is profitability; only the q5 model spans both factors (GRS $p = 0.32$).

4 Combining adds diversification, not return: as long-short factors they sit below long-only equity benchmarks.

Note: benchmarks are long-only total-return indices (excess of RF); Z / F / Combined are long-short, zero-cost. MSCI Quality > S&P reflects this, so Sharpe ratios are not directly comparable.

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Data: Global Factor Data (jpkpfactors.com), global-q.org, Ken French Data Library, Bloomberg, FRED.